

Integrated Robotic Actuator Technical Design Document

Electromechanical architecture, control design, verification, and integration plan

PROJECT

Integrated Robotic Actuator

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Abstract

This document details the design development of the Node actuator mechanical housing, integrated with the cycloidal reducer, based on requirements from NT-N-OVR-000, and research findings from NT-N-MH-000 and NT-N-CR-000. It includes derivation of design constraint regions based on requirements, and the final design solution with justification.

Keywords: robotic actuator, BLDC, embedded control, torque sensing, mechatronics

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REVISION HISTORY

Rev.	Date	Author	Change Description
A	2026-06-25	NervTech Engineering	Initial technical document class demonstration.

1 SYSTEM OVERVIEW

System Summary

Architecture	Adhesive bonded stator and rotor assembly with integrated cycloidal reducer and space for embedded control electronics.
Control	Integrated PCB for FOC control, communication and power supply via USB-C or Molex.
Sensing	Hall effect sensors for rotor position and speed feedback, with additional quadrature encoder for higher resolution.
Safety	Low reduction ratio for high backdrivability, and vibration compensation in cycloidal drive for less transfer to fixture.

2 REQUIREMENTS

There are several goals for Node that drive the requirements and design of the actuator.

2.1 NT-N-G1

From NT-N-OVR-000, NT-N-G1 has the following goal and success criteria:

- Goal: "The actuator should provide high precision torque and position control in high motion frequency applications."
- Success Criteria: "The actuator should be able to move standard payloads up to 20kgs at frequencies comparable to human motion."

The components chosen must be able to account for failure modes induced by the stresses produced by the output load of up to 200N, applied at lever arms up to 0.25m, at frequency bandwidths up to 20Hz. This interpretation of "frequencies comparable to human motion" is derived from [1], which shows that human motion is typically in the range of 0.5Hz to 20Hz, however extended use of the upper limit is rarely reached. The lever arm of 0.25m is derived from the half length of a human arm, which is approximately 0.6m, with the 200N payload derived from the weight of a 20kg object, which would be approximately 196N.

The cycloidal speed reducer has inherent vibration characteristics, which must be accounted for in the design of the housing to ensure that the vibrations are not transferred to the fixture. Since the cycloidal speed reducer also has inherent hertzian contact stresses, the housing must be able to withstand these stresses without failure.

For the high accuracy and precision postional requirement, the overall backlash in the system should be kept to a maximum of 4 arcminutes, which would give a maximum positional error of 0.2525mm at a 0.25m lever arm, which is acceptable for the intended applications of the actuator.

Torque ripple must also be kept to a minimum to meet the high accuracy and precision torque requirement, which is defined as a maximum of 0.5% of the rated torque, which is 0.25Nm for the actuator.

2.2 NT-N-G2

From NT-N-OVR-000, NT-N-G2 has the following goal and success criteria:

Goal: "The actuator should be reusable and easily integratable into a variety of frames for robotic systems."

Success Criteria: "The actuator should be able to be mounted and dismounted with conventional tools from any of the standard NervTech linkages and/or frames."

The housing should be able to be mounted via standardised metric machine screws, with the fasteners being able to be tightened with a standard hex key or torx key. The fasteners must also be able to withstand the maximum torque applied to the actuator, which is 50Nm, without loosening or failing.

The output flange must have the same mounting pattern as the mounting base, as this pattern will provide the basis for all future linkages and actuators in the Node family of products, and will allow for easy integration of the actuator into a variety of frames for robotic systems.

2.3 NT-N-G7

From NT-N-OVR-000, NT-N-G7 has the following goal and success criteria:

Goal: "The actuator should be durable and reliable for long term use."

Success Criteria: "The actuator should be able to operate for at least 10,000 hours of continuous operation without failure, and should be able to withstand standard industrial environmental conditions including dust and high moisture."

The housing will be exposed to large temperature variations from stator heating, and as such it must be able to manage this heat long term to prevent failure. The housing will also be highly vulnerable to failure from metallic dust and contaminants unless sealed to an appropriate IP rating, which will be determined based on the final design solution. This may prevent a hollow shaft design, as this would require a large opening in the housing to allow for the shaft to pass through, which would compromise the sealing of the housing.

Since cycloidal speed reducers have inherent hertzian contact stresses, the material selection for the cycloidal plates and rollers must be appropriate to prevent failure from these stresses, and the housing must be able to withstand the forces produced by these stresses without failure (including increase in backlash and torque ripple over time). Cycloidal reducer bearings must also be chosen to withstand the forces produced by the output load at the maximum frequency of 20Hz, and the housing must be able to withstand the forces produced by these bearings without failure.

Table 1: Requirements

ID	Type	Requirement	Driving Goal	Verification
NT-N-IRQ1	Performance	The actuator shall withstand an output load of up to 200N applied at a lever arm of up to 0.25m at motion frequencies up to 20Hz without mechanical failure, including creep, plastic deformation, and fatigue.	NT-N-G1	Structural analysis and dynamic load test
NT-N-IRQ2	Vibration	The actuator housing shall limit the transfer of vibration generated by the cycloidal speed reducer to the external fixture.	NT-N-G1	Vibration test; acceptance criterion TBD
NT-N-IRQ3	Structural	The actuator housing shall withstand the reaction forces generated by Hertzian contact within the cycloidal speed reducer without structural failure.	NT-N-G1	Stress analysis and load test
NT-N-IRQ4	Precision	The total mechanical backlash of the actuator shall not exceed 4 arcminutes.	NT-N-G1	Backlash measurement test
NT-N-IRQ5	Precision	The actuator torque ripple shall not exceed 0.5% of rated torque, corresponding to 0.25Nm at the 50Nm rated torque.	NT-N-G1	Torque ripple measurement
NT-N-IRQ6	Interface	The actuator housing shall be mountable and removable using standard metric machine screws tightened using conventional hex or Torx tools.	NT-N-G2	Inspection and assembly test
NT-N-IRQ7	Structural	The actuator mounting fasteners shall withstand the maximum actuator torque of 50Nm without loosening or mechanical failure.	NT-N-G2	Fastener analysis and proof-load test
NT-N-IRQ8	Interface	The actuator output flange shall use the same mounting pattern as the actuator mounting base.	NT-N-G2	Dimensional inspection
NT-N-IRQ9	Reliability	The actuator shall be capable of at least 10,000 hours of continuous operation without failure.	NT-N-G7	Reliability analysis and endurance testing
NT-N-IRQ10	Environmental	The actuator housing shall protect internal components from dust and high-moisture industrial environments using an appropriate enclosure sealing level.	NT-N-G7	IP environmental test; required rating TBD

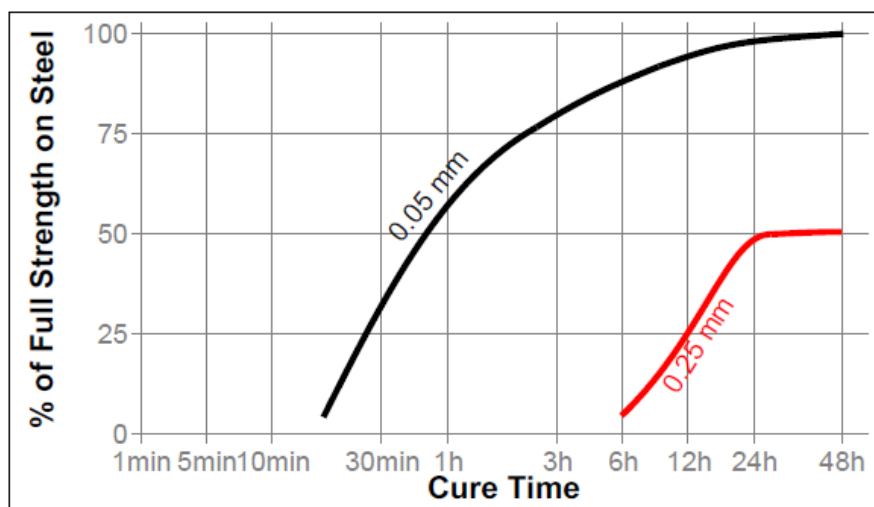
Table 2: Requirements

ID	Type	Requirement	Driving Goal	Verification
NT-N-IRQ11	Thermal	The actuator housing shall manage heat generated by the stator during continuous operation such that thermal loading does not cause component or housing failure.	NT-N-G7	Thermal analysis and continuous-load thermal test
NT-N-IRQ12	Durability	The cycloidal plates and roller pins shall use materials capable of withstanding the expected Hertzian contact stresses over the required operating life without failure.	NT-N-G7	Hertzian stress analysis and material verification
NT-N-IRQ13	Durability	The cycloidal reducer bearings shall withstand the forces generated by the maximum specified output load at operating frequencies up to 20Hz without failure.	NT-N-G7	Bearing load and life analysis
NT-N-IRQ14	Structural	The actuator housing shall withstand the reaction forces transmitted by the cycloidal reducer bearings throughout the required operating life without structural failure.	NT-N-G7	Structural analysis and endurance test
NT-N-IRQ15	Durability	Wear and deformation of the cycloidal reducer and its supporting structure shall not cause backlash or torque ripple to exceed their specified limits over the required operating life.	NT-N-G7	Endurance test with pre- and post-test backlash and torque-ripple measurement

2.4 Rotor Cup Design

2.4.1 Bond Line Specifications

The rotor cup design must be mounted to the rotor such that the rotor cup does not slip or rotate relative to the rotor during operation, as this would cause a loss of torque transmission and potentially damage the actuator (NT-N-IRQ1). From research done in NT-N-MH-000, adhesive bonding using retaining compounds will be the preferred method to accomplish this goal. The following retaining compound has been identified as a suitable candidate for this application:

**Figure 1:** Bond Line Thickness vs. Cure Strength for Retaining Compounds

From the data in Figure 1, it can be seen that the bond line thickness must be kept to a minimum of 0.05mm, and a maximum of 0.25mm, to ensure the bond line strength is between 50% and 100%, assuming steel on steel bonding. The closer the bond line thickness is to 0.05mm, the higher the bond line strength will be, and the more torque can be transmitted through the bond line without failure. From the WK8115 datasheet specification (see NT-N-MH-000), the tolerance of the rotor back iron outer diameter is ± 0.1 mm, which already exceeds the minimum bond line thickness of 0.05mm.

2.4.2 Rotor Cup Tolerances

The rotor cup tolerance must be within ± 0.025 mm to ensure that the bond line thickness has a maximum thickness of 0.25mm, which in the worst case reduces the bond line strength to 50% of the maximum strength on steel. This is suitable for the application, as the maximum torque transmitted through the bond line is 5.83Nm from NT-N-MH-000, and the breakaway torque at 50% is at a minimum 10Nm after curing.

It must also be located such that its central axis is aligned with the rotor axis, as any misalignment would cause uneven air gaps (see NT-N-MH-000 for preliminary research), leading to torque ripple and potential damage to the actuator (NT-N-IRQ1) (NT-N-IRQ5). Slots for locating pins can be machined onto the backiron at 180° separation, and corresponding slots can be machined onto the rotor cup to accommodate these pins (NT-N-IRQ1) (NT-N-IRQ5).

2.4.3 Rotor Cup Cycloidal Integration

From the research done in NT-N-CR-000, to integrate an epitrochoidal-pinwheel speed reducer into the actuator (NT-N-IRQ1), 2 epitrochoidal profiles in 180° phase separation must be machined onto the output side of the rotor cup. The 2 epitrochoidal profiles reduce the vibration from the eccentric orbit of one profile (NT-N-IRQ2). To directly machine the epitrochoidal profiles into the rotor cup, the rotor cup must have bearings that can withstand the overturning moment forces of the cycloidal output, and the radial forces from their respective eccentric orbits (NT-N-IRQ15).

2.4.4 Rotor Cup Stress Analysis

There should be no varying axial forces on the output flange of the actuator. According to NT-N-IRQ1, the actuator should be able to apply 200Nm at a lever arm of 0.25m, up to 50Hz. Resisting axial forces as a result of moving loads is not mentioned in the requirements and is therefore not considered in the stress analysis. Static axial forces will, however, be considered as the actuator may be mounted in a variety of configurations, which may include resistance of a payload's weight when mounted parallel to the ground.

REFERENCES

- [1] Piyarathna, Iresha & Thabet, Ahmed & Ucgul, Mustafa & Lemckert, Charles & Lim, Yee Yan & Tang, Zi Sheng. (2023). Linear Segmented Arc-Shaped Piezoelectric Branch Beam Energy Harvester for Ultra-Low Frequency Vibrations. Sensors. 23. 10.3390/s23115257.